

QUERYING DATA

1. DISTINCT VALUES

```
1 • use employee;  
2 • select distinct salary from employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

salary
60000.00
80000.00
45000.00
70000.00
55000.00
65000.00
90000.00
95000.00
75000.00
85000.00
100000.00

Result Grid
Form Editor
Field Types
Query Stats

2. ALIAS(AS)

```
1 • use employee;  
2 • select distinct salary from employees;  
3 • select age as employee_age, salary as employee_salary from employees;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

employee_age	employee_salary
27	60000.00
31	80000.00
26	45000.00
29	70000.00
25	55000.00
28	65000.00
32	90000.00
27	70000.00
30	55000.00
29	80000.00
28	60000.00
26	95000.00
33	75000.00
30	60000.00
27	85000.00

Result Grid
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Query Stats

3. WHERE CLAUSE & OPERATORS

```
1 • use employees;
2 • select distinct salary from employees;
3 • select age as employee_age, salary as employee_salary from employees;
4 • select * from employees
5   where salary > 50000 and hire_date < '2016-01-01';
6
7 • select * from employees where designation is null;
```

Result Grid

	EMPLOYEE_ID	EMPLOYEE_NAME	GENDER	AGE	HIRE_DATE	DESIGNATION	DEPARTMENT_ID	LOCATION_ID	SALARY
▶	5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000.00
	5002	Reyansh Singh	M	31	2015-03-10	Network Engineer	12	1	80000.00
	5004	Kiara Malhotra	F	29	2015-07-05	data scientist	8	3	70000.00
	5005	Anvi Chaudhary	F	25	2015-09-11	Business Development Executive	11	1	55000.00
	5006	Dhruv Shetty	M	28	2015-11-20	UI Developer	8	2	65000.00
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Form Editor
Field Types
Query Stats

SORTING AND GROUPING DATA

1. ORDER BY

```
8
9 • update employees
10   set designation = 'Data scientist'
11   where designation is null;
12
13 • select * from employees
14   order by department_id asc, salary desc;
```

Result Grid

	EMPLOYEE_ID	EMPLOYEE_NAME	GENDER	AGE	HIRE_DATE	DESIGNATION	DEPARTMENT_ID	LOCATION_ID	SALARY
▶	5020	Aarav Nair	M	32	2018-03-05	Software Engineer	1	1	90000.00
	5007	Anushka Singh	F	32	2016-01-15	Marketing Manager	2	3	90000.00
	5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	2	2	70000.00
	5011	Saanvi Patel	F	28	2016-09-20	Marketing Analyst	2	1	60000.00
	5025	Avni Iyengar	F	31	2019-01-20	Data Scientist	3	4	100000.00
	5026	Vivaan Singh	M	29	2019-03-10	Business Analyst	3	2	75000.00
	5001	Vihaan Singh	M	27	2015-01-20	Data Analyst	3	4	60000.00
	5016	Zoya Khan	F	31	2017-07-05	Legal Counsel	4	4	100000.00
	5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	4	4	60000.00
	5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	4	4	55000.00
	5024	Aryan Shetty	M	27	2018-11-05	Product Manager	5	1	95000.00
	5030	Sara Iyer	F	28	2019-11-20	Project Manager	5	1	90000.00
	5027	Ananya Paul	F	32	2019-05-20	Content Writer	6	3	60000.00
	5028	Anaya Kapoor	F	26	2019-07-05	Event Coordinator	6	1	60000.00
	5015	Ishaan Kumar	M	27	2017-05-20	Financial Analyst	7	1	85000.00

Form Editor
Field Types
Query Stats

EMPLOYEE_ID	EMPLOYEE_NAME	GENDER	AGE	HIRE_DATE	DESIGNATION	DEPARTMENT_ID	LOCATION_ID	SALARY
5016	Zoya Khan	F	31	2017-07-05	Legal Counsel	4	4	100000.00
5025	Avni Iyengar	F	31	2019-01-20	Data Scientist	3	4	100000.00
5012	Myra Verma	F	26	2016-11-05	Operations Manager	13	2	95000.00
5024	Aryan Shetty	M	27	2018-11-05	Product Manager	5	1	95000.00
5007	Anushka Singh	F	32	2016-01-15	Marketing Manager	2	3	90000.00
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	1	1	90000.00
5030	Sara Iyer	F	28	2019-11-20	Project Manager	5	1	90000.00
5015	Ishaan Kumar	M	27	2017-05-20	Financial Analyst	7	1	85000.00
5021	Adivik Kapoor	M	26	2018-05-20	Finance Analyst	7	3	85000.00
5002	Reyansh Singh	M	31	2015-03-10	Network Engineer	12	1	80000.00
5010	Atharv Yadav	M	29	2016-07-10	Systems Administrator	12	4	80000.00
5017	Kabir Nair	M	28	2017-09-11	IT Support Specialist	12	2	80000.00
5029	Arjun Kumar	M	33	2019-09-11	Quality Assurance ...	12	2	80000.00
5013	Arnab Rao	M	33	2017-01-20	Customer Success ...	10	3	75000.00
5018	Ishan Mishra	M	25	2017-11-20	Research Scientist	9	3	75000.00

```
13 • select * from employees  
14     order by department_id asc, salary desc;  
  
15  
16 • select * from employees  
17     where year(hire_date)=2018  
18     order by hire_date  
19     limit 5;
```

Result Grid

EMPLOYEE_ID	EMPLOYEE_NAME	GENDER	AGE	HIRE_DATE	DESIGNATION	DEPARTMENT_ID	LOCATION_ID	SALARY
5019	Ishika Patel	F	29	2018-01-15	Talent Acquisition Specialist	4	4	55000.00
5020	Aarav Nair	M	32	2018-03-05	Software Engineer	1	1	90000.00
5021	Advik Kapoor	M	26	2018-05-20	Finance Analyst	7	3	85000.00
5022	Aadhya Iyengar	F	28	2018-07-10	HR Specialist	4	4	60000.00
5023	Anika Paul	F	30	2018-09-20	Public Relations Specialist	2	2	70000.00
* NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Form Editor
Field Types
Query Stats

3. AGGREGATE FUNCTIONS

```
18  order by hire_date
19  limit 5;
20
21  • SELECT SUM(E.SALARY) AS Total_Finance_Salary
22  FROM EMPLOYEES E
23  JOIN DEPARTMENTS D ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
24  WHERE D.DEPARTMENT_NAME = 'Finance';
25
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Total_Finance_Salary
170000.00

```
21  • SELECT SUM(E.SALARY) AS Total_Finance_Salary
22  FROM EMPLOYEES E
23  JOIN DEPARTMENTS D
24  ON E.DEPARTMENT_ID = D.DEPARTMENT_ID
25  WHERE D.DEPARTMENT_NAME = 'Finance';
26
27  • select min(age) as Minimum_Age from employees;
28
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Minimum_Age
25

4. GROUP BY

```
26
27 • select min(age) as Minimum_Age from employees;
28
29 • select l.location, max(e.salary) as max_salary
30 from employees e
31 join location l on e.location_id = l.location_id
32 group by l.location;
```

location	max_salary
Bangalore	95000.00
Chennai	95000.00
Hyderabad	90000.00
Pune	100000.00

```
31 join location l on e.location_id = l.location_id
32 group by l.location;
33
34 • select designation, avg(salary) as Average_Salary
35 from employees
36 where designation like '%Analyst%'
37 group by designation;
```

designation	Average_Salary
Data Analyst	60000.000000
Marketing Analyst	60000.000000
Supply Chain Analyst	60000.000000
Financial Analyst	85000.000000
Finance Analyst	85000.000000
Business Analyst	75000.000000
Quality Assurance Analyst	80000.000000

5. HAVING

```
37     group by designation;
38
39 • select d.department_name,count(e.employee_id) as Employee_Count
40     from departments d
41     left join employees e on d.department_id=e.department_id
42     group by d.department_id,d.department_name
43     having count(e.employee_id)<3;
44
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
department_name	Employee_Count		
Business Development	2		
Content Creation	2		
Finance	2		
Operations	1		
Product Management	2		
Research and Development	1		
Software Development	1		

```
43     having count(e.employee_id)<3;
44
45 • select l.location,avg(e.age) as Average_Age from employees e
46     join location l on e.location_id = l.location_id
47     where e.gender = 'f'
48     group by l.location
49     having avg(e.age) < 30;
50
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
location	Average_Age		
Bangalore	27.3333		
Chennai	26.7500		
Pune	29.2000		

JOINS

1. INNER JOIN

```
47 where e.gender = 'f'
48 group by l.location
49 having avg(e.age) < 30;
50
51 • SELECT E.EMPLOYEE_NAME, E.DESIGNATION, D.DEPARTMENT_NAME
52 FROM EMPLOYEES E
53 INNER JOIN DEPARTMENTS D ON E.DEPARTMENT_ID = D.DEPARTMENT_ID;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
EMPLOYEE_NAME	DESIGNATION	DEPARTMENT_NAME	
Anvi Chaudhary	Business Development Executive	Business Development	
Kiaan Desai	Sales Executive	Business Development	
Ananya Paul	Content Writer	Content Creation	
Anaya Kapoor	Event Coordinator	Content Creation	
Aaradhya Iyer	Customer Support Executive	Customer Support	
Arnav Rao	Customer Success Manager	Customer Support	
Vihaan Mohan	Supply Chain Analyst	Customer Support	
Vihaan Singh	Data Analyst	Data Science	
Avni Iyengar	Data Scientist	Data Science	
Vivaan Singh	Business Analyst	Data Science	
Kiara Malhotra	data scientist	Design	
Dhruv Shetty	UI Developer	Design	
Diya Jha	Graphic Designer	Design	
Ishaan Kumar	Financial Analyst	Finance	
Advik Kapoor	Finance Analyst	Finance	

2. LEFT JOIN

```
52 FROM EMPLOYEES E
53 INNER JOIN DEPARTMENTS D ON E.DEPARTMENT_ID = D.DEPARTMENT_ID;
54
55 • SELECT D.DEPARTMENT_NAME, COUNT(E.EMPLOYEE_ID) AS Total_Employees
56 FROM DEPARTMENTS D
57 LEFT JOIN EMPLOYEES E ON D.DEPARTMENT_ID = E.DEPARTMENT_ID
58 GROUP BY D.DEPARTMENT_ID, D.DEPARTMENT_NAME;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
DEPARTMENT_NAME	Total_Employees		
Business Development	2		
Content Creation	2		
Customer Support	3		
Data Science	3		
Design	3		
Finance	2		
Human Resources	3		
IT	4		
Marketing	3		
Operations	1		
Product Management	2		
Research and Develo...	1		
Software Development	1		

3. RIGHT JOIN

```
56 FROM DEPARTMENTS D
57 LEFT JOIN EMPLOYEES E ON D.DEPARTMENT_ID = E.DEPARTMENT_ID
58 GROUP BY D.DEPARTMENT_ID, D.DEPARTMENT_NAME;
59
60 • SELECT L.LOCATION, E.EMPLOYEE_NAME
61 FROM EMPLOYEES E
62 RIGHT JOIN LOCATION L ON E.LOCATION_ID = L.LOCATION_ID;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
LOCATION	EMPLOYEE_NAME		
Bangalore	Aaradhya Iyer		
Bangalore	Dhruv Shetty		
Bangalore	Myra Verma		
Bangalore	Vihaan Mohan		
Bangalore	Kabir Nair		
Bangalore	Anika Paul		
Bangalore	Vivaan Singh		
Bangalore	Arjun Kumar		
Chennai	Reyansh Singh		
Chennai	Anvi Chaudhary		
Chennai	Saanvi Patel		
Chennai	Ishaan Kumar		
Chennai	Aarav Nair		
Chennai	Aryan Shetty		
Chennai	Anaya Kapoor		